# Diseases of marigold

1. **Cercospora leaf spot and blight** - ***Cercospora tagetis, C. tageticola, C. thunbergiae***

2. **Colletotrichum leaf spot /Anthracnose** - ***Colletotrichum* spp**

3. **Alternaria leaf spot and flower blight** - ***Alternaria tagetica, A. zinniae, A. dianthi***

4. **Powdery mildew** - ***Oidium* sp (*Leveillula taurica*)**

5. **Flower blight** - ***Botrytis cinerea***

# Cercospora leaf spot and blight - *Cercospora tagetis, C. tageticola, C. thunbergiae,*

# **Cercospora leaf spot and blight - *Cercospora tagetis, C. tageticola, C. thunbergiae***

## Symptoms

***Cercospora tageticola*:** Spots are circular, dark brown with **prominent grey margin with effuse growth.**

***C. tagetis* :** Spots are **amphigenous, scattered, irregular /indefinite, blackish, coalescent.**

***C. thunbergiae* :** Spots are circular to oblong, brown **with blackish brown margins**

# Cercospora leaf spot / blight - *Cercospora tagetis, C. tageticola, C. thunbergiae*

**Pathogen:**

***C. tageticola*** Fruiting is effuse.
***C. tagetis –*** stromata olivaceous-brown, small, globular to iIrregular on the lower side of leaf

- Pathogen survives mainly in **plant debris as desiccation-resistant pseudostromata**, but **can also survive as conidia in debris or seeds.**
- When moisture is sufficient, new conidia are formed and spread via rain-splash or wind to new leaves or plants.
- Optimum conditions are 25° to 35° C with night temperatures above 16.1° C, and relative humidity of 90-95%.
- **Infection is greatly reduced or nonexistent** whenever the leaf wetness is less than 11 hours or at temperatures less than 15° C.

# Colletotrichum leaf spot - *Colletotrichum* spp

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## Symptoms

- **Small, circular lesions on leaves and petals** which coalesce to form large areas of necrotic tissue.
- **Leaves and petals may become completely** blighted.
- A diagnostic sign is **the presence of pink to orange sporulation oozing** from lesions.

# Alternaria leaf spot and flower blight - *Alternaria tagetica*

**Alternaria leaf and stem spot.**

# Alternaria leaf spot and flower blight - *Alternaria tagetica*

# Alternaria leaf spot and blight - *Alternaria tagetica*

## Symptoms

- *Alternaria tagetica* can result in **blight on leaves, stems, and flowers**.
- Dark-brown necrotic spots form initially and may be surrounded by a chlorotic halo.
- **Leaf fall may take place and flower size** is reduced
- When the disease attains serious proportions, plants show blighted appearance.
- On the **stem lesions or cankers** may develop
- **Flower petals** turn totally **dark and shrivelled**

# Leaf spot and flower blight - *Alternaria zinniae, / A. dianthi*

# Leaf spot and flower blight - *Alternaria zinniae, / A. dianthi*

## Symptoms

- **Disease appears mainly on young flower buds** and results into their dry rotting with brown scorched symptoms
- Necrotic discoloration of sepals and stalk
- The ray and disc florets also turn brown.
- At later stages buds become shrivelled, turn dark brown and dry up.
- Symptoms are less prominent on mature buds but these buds fail to open
- Severe infection of **flower blight** causes reduced seed yield, seed viability and flower quality
- The pathogen **also infects leaves causing blight.**
- **Brown necrotic spots are visible on margins** and **tips of older leaves**

# Powdery mildew - *Oidium* sp.; *Leveillula taurica*

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## Symptoms

- **Frosty white, powdery spots** develop on leaves.
- Colonies can become so extensive that the entire plant may appear white.

# Powdery mildew - *Oidium* sp.; *Leveillula taurica*

## Management

- Space plants for good air circulation.
- Remove and destroy infected plants.
- Plant in sunny open locations rather than in shady areas.

# Botrytis flower blight of marigold - *Botrytis cinerea*

Gray Mold on Marigold leaves

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# Botrytis flower blight of marigold - *Botrytis cinerea*

## Symptoms

- **Oval to irregular, gray to black spots with tiny dots** peppering their surface (fruiting structures) form first on **lower leaves and then spread upward**.
- Tan to brown spots with a target-like appearance can also develop on the leaves.
- **Flower petals are most susceptible** at any stage of development **either all petals in flower or a few** may become diseased.
- Flower parts develop water-soaked lesions that become necrotic and die.
- A gray mass of spores develops on necrotic tissue during wet conditions.
- **Senescing flowers** are particularly susceptible.
- **Senescing leaves** may also be infected.

# Botrytis flower blight of marigold - *Botrytis cinerea*

## Symptoms contd…

- Infection of blossoms by *Botrytis* may cause **premature fading and dying of petals.**

- These patches are often associated with flowers which have dropped onto the leaf surface.

- This disease is particularly problem during periods of **extended cloudy, humid, wet weather.**

# Botrytis flower blight of marigold - *Botrytis cinerea*

## Pathogen

- The fungus that colonizes **dead, dying, and wounded plant parts** and can attack healthy tissues.
- **A moist, humid environment** is ideal for pathogen sporulation and spread.
- Conidia may come from sources in and outside the greenhouse.
- **Wide host range**.

# Botrytis flower blight of marigold - *Botrytis cinerea*

## Management

- Remove **all dead and dying plant parts** (particularly blossoms) on the plants.
- **Avoid overhead irrigation o**r apply such that plants are not wet for extended periods of time.
- Maintain proper spacing for good air circulation.
- Apply 0.2 % Chlorothalonil or 0.1 % Azoxystrobin or Fludioxonil or Thiophanate methyl

# Diseases of tuberose

1. **Alternaria leaf spot** - ***Alternaria polianthi***
2. **Macrophoma leaf spot** -

# Alternaria leaf spot - *A. polianthi*

**Fig. 1: Symptoms of Leaf spot of Tuberose (*Alternaria polyanthi*)**

# Alternaria leaf spot - *A. polianthi*

*A. polyanthi*

# Alternaria leaf spot - *Alternaria polianthi*

## Symptoms

- **Leaf spots** first appear during the **flower stalk initiation stage** and increased gradually with the advancement of crop growth.
- **Faint concentric rings appear on mid rib of leaves but rarely near the margin of leaves**
- **The leaves become necrotic and dry up with the infection**
- **Occasionally peduncle** may also be infected showing circular to oval spots which measure 10-30 mm in length and 4-5 mm in diameter.
- **Infected leaves and peduncles** become necrotic and dry up

# Macrophoma leaf spot - *Macrophoma*